BIKE SHARING DEMAND ANALYSIS

Excel-Based Analysis, Dashboard and Basic Automation

Project Summary Report

1. Project Overview

This project uses Microsoft Excel to explore bike-sharing rental demand and present the results in a clear, business-friendly format. The workbook combines rental and contextual fields such as date, hour, weather, holiday status and user type. The goal is to summarize patterns that can support operational planning.

2. Data and Workbook Structure

The workbook contains source-data sheets (dataset_1, dataset_2 and dataset_3), four analysis sheets (Sheet1–Sheet4), a BUSINESS INSIGHTS sheet, a DASHBOARD sheet and a FORECAST sheet. The main combined dataset has 1,000 data rows plus a header row. It includes time-related fields, weather measures and rental counts.

The workbook includes lookup formulas and duplicate/check columns. The visible checks indicate that records were checked for duplication and consistency. Before reusing the workbook with new data, the full input range and formulas should be reviewed to ensure the checks cover all imported rows.

3. Excel Methods Used

- Data organization and worksheet structuring for the source tables and combined dataset.
- Lookup formulas to bring related fields into the main analysis dataset.
- Summary calculations and statistical measures, including totals, proportions, standard deviation and correlations.
- Pivot Tables and Pivot Charts to compare rental demand by hour, weather situation, holiday status and user type.
- A forecast worksheet with daily rental summaries and Excel linear-forecast formulas.

4. Dashboard and Key Findings

The dashboard presents a total-rentals KPI, registered-versus-casual user shares and charts for hourly demand, weather situations, holiday versus non-holiday demand, and casual versus registered users by day of week.

Metric	Workbook result
Total rentals shown in Business Insights	58,304
Registered-user share	91.56%
Casual-user share	8.44%
Temperature vs rentals correlation	0.20 (weak positive relationship)
Humidity vs rentals correlation	-0.25 (weak negative relationship)
Windspeed vs rentals correlation	0.10 (very weak positive relationship)
Standard deviation of rental counts	50.99

Based on the values recorded in the workbook, registered users account for the majority of rentals. The listed weather correlations are relatively weak, so they should be interpreted as descriptive signals, not proof that weather alone causes changes in demand.

5. Forecasting and Automation

The FORECAST worksheet contains daily rental totals, forecast dates and predicted rentals using Excel's linear forecasting function. These outputs should be treated as a basic demonstration rather than a validated prediction model. Forecast quality depends on the dates and source range used.

A VBA procedure named Refresh_Bike_Sharing_Dashboard was added. It turns off screen updating, calls ThisWorkbook.RefreshAll, activates the DASHBOARD sheet, selects cell A1 and displays a completion message. The macro ran and displayed its message. That message confirms the procedure reached its message step; it does not independently prove that every external query or data connection refreshed successfully.

6. Recommendations

- Use the dashboard to review hourly and user-type patterns before planning staffing or bike availability.
- Keep the source data, formulas and chart ranges consistent when importing new records.
- Refresh the workbook and verify query status and updated values before sharing a new report.
- Treat the forecast as an indicative trend until it is tested against additional historical data.
- For future improvements, add a documented anomaly check and test the workbook with a fresh data import.

7. Conclusion

The workbook demonstrates a practical Excel workflow: organizing data, combining related fields, summarizing demand with Pivot Tables and charts, presenting KPIs in a dashboard, producing a basic linear forecast and adding a simple VBA refresh procedure. The main business takeaway from the current workbook is that registered users contribute most of the rentals. Results should be interpreted in the context of the current dataset size and verified again whenever the source data is updated.

Prepared for the Bike Sharing Demand Analysis Excel Project